

MORAN: A Multi-Object Rectified Attention Network for Scene Text Recognition

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Abstract

Irregular text is widely used. However, it is considerably difficult to recognize because of its various shapes and distorted patterns. In this paper, we thus propose a **multi-object rectified attention network (MORAN)** for general scene text recognition. The MORAN consists of a multi-object rectification network and an attention-based sequence recognition network. The multi-object rectification network is designed for rectifying images that contain irregular text. It decreases the difficulty of recognition and enables the attention-based sequence recognition network to more easily read irregular text. It is trained in a weak supervision way, thus requiring only images and corresponding text labels. The attention-based sequence recognition network focuses on target characters and sequentially outputs the predictions. Moreover, to improve the sensitivity of the attention-based sequence recognition network, a fractional pickup method is proposed for an attention-based decoder in the training phase. With the rectification mechanism, the MORAN can read both regular and irregular scene text. Extensive experiments on various benchmarks are conducted, which show that the MORAN achieves state-of-the-art performance. The source code is available¹.

Keywords: Scene text recognition, optical character recognition, deep learning.



Figure 1. Examples of regular and irregular scene text. (a) Regular text. (b) Slanted and perspective text. (c) Curved text.

1. Introduction

Scene text recognition is an essential process in computer vision tasks. Many practical applications such as traffic sign reading, product recognition, intelligent inspection, and image searching, benefit from the rich semantic information of scene text. With the development of scene text detection methods [11, 26, 46, 56], scene character recognition has emerged at the forefront of this research topic and is regarded as an open and very challenging research problem [45].

Nowadays, regular text recognition methods [3, 33, 41, 45, 50] have achieved notable success. Moreover, methods based on convolutional neural networks [3, 22, 50] have been broadly applied. Integrating recognition models with recurrent neural networks [17, 41, 42] and attention mechanisms [5, 6, 27, 51] yields better performance for these models.

Nevertheless, most current recognition models remain too unstable to handle multiple disturbances

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¹https://github.com/Canjie-Luo/MORAN_v2